

Computer Vision Project Report
Submitted by:

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Abstract

This project intends to meet the important demand for early and precise detection of plant diseases, which has a substantial influence on agricultural output and sustainability. The project's goal is to create a system that can classify plant diseases from photos using computer vision and deep learning. The goal is to properly preprocess plant photos and utilize advanced neural networks, including custom models and the VGG16 pretrained model, to achieve reliable and scalable disease detection. This contribution lays the groundwork for automated disease detection, allowing for more prompt intervention and better agricultural outcomes.

1. Related Work

An advanced deep convolutional neural network (Deep CNN) was put forward in this paper (Geetharamani & Arun Pandian, 2019) for diagnosing plant leaf diseases using a dataset containing 39 different classes of plant leaves, including both health and disease portions. In contrast to more conventional machine learning models where feature extraction is typically done manually, Deep CNN adopted an end-to-end learning, which essentially enhanced the models' classification capabilities. The authors concisely illustrated using flipping, gamma correction, noise injection, PCA color augmentation, rotation, and scaling that how incorporating these data augmentation techniques helped in increasing the size of dataset and improving model generalization capability. The last model, M4-Both, was tested, and the average accuracy reached was 96.46 % and is higher than other transfer learning techniques.

In this study (Rao et al, 2021), an application of transfer learning using Alex Net for diseases detection in plants especially grape and mango leaves is shown. Thus, developed using the dataset containing 8,438 images including 7,222 Plant Village images and 1,266 new mango leave images, accuracy of 99.03% for the grape and 89% for mango leave was observed for developed model. The system developed was using MATLAB and extended under the "JITCROPFIX" mobile application allowing farmers to diagnose diseases and reduce losses economically. The work's main contribution is emphasizing the prospects of deploying pre-trained CNN models to address the above-mentioned problems in agriculture, especially in countries such as India, where the lack of development prevents the rapid identification of diseases. However, it also admits the following key restrictions: data paucity from different geographical and lighting conditions, along with complications in detecting minor deformities at an early stage. Future directions involve using such a model for other

diseases; putting in place a recommendation system for disease management; and integrating the system in self-driving drones for field surveillance. These developments underscore how AI transforms precision farming with focus on cost leadership and scalability of disease prevention.

This paper introduces a new deep learning-based object detection model aimed at multi-class plant disease detection, with focus on apple plants. The model uses an improved YOLOv4 configuration pre-trained on the GitHub's COCO dataset and fine-tuned for a higher computing speed and effectiveness. To overcome the issue of detection at multiple scales and with high fineness for the disease in question, the proposed model is quite effective, regretfully with a mAP of 0.912, F1-score of 0.959, and at an FPS of 56.9 (Roy & Bahduri, 2021) . This model is particularly useful for real-world and complex orchard conditions where many previous models fail in small scale and early stage of disease. The proposed system is better than prior models, obtaining a 9.05% higher precision as well as 7.6% higher F1-score, that underlines its effectiveness in real-time disease diagnosis. Additionally, the research also proposes that the model can be applied to other agricultural areas such as disease discovery in various crops, the development of automatic monitoring systems, and large-scalable solutions for precision farming. It is understood from this paper that the increased efficiency of disease detection using this model will lead to prevention management, which is likely to have huge impacts economically and environmentally.

A recent study utilizing the same dataset as this work evaluated various transfer learning models for classifying 38 plant disease classes. Among the tested architectures, DenseNet-121 outperformed ResNet-50, VGG-16, and Inception V4, achieving a remarkable classification accuracy of 99.81% and an F1 score of 99.8%. The authors highlighted DenseNet-121's efficiency, attributing its superior performance to a smaller number of trainable parameters and reduced computational complexity, making it particularly suitable for extending to new plant diseases (Eunice et al., 2022).

The identification of diseases in plant crops for instance olive tree diseases is very essential in management of disease in crops especially in areas such as Saudi Arabia (Ksibi, Ayadi, Soufiene, Jamjoom, & Ullah, 2022). The previous practices are slow and unreliable; hence, advances in deep learning (DL) make the detection of diseases much more accurate and automatic. A new model called MobiRes-Net which is formed by integrating ResNet50 and MobileNet using deep feature concatenation was deployed to improve disease detection in olive leaves. The proposed model here works on the images captured

Based on the findings of authors Bagga & Goyal in 2023, deep learning (DL) models, notably Convolutional Neural Networks (CNNs), have made considerable advances in plant disease diagnosis, with accuracies ranging from 85% to over 90%. Researchers improved model performance by using preprocessing approaches such as picture scaling, noise filtering, and contrast enhancement. Transfer learning with pre-trained models, such as ResNet, Inception, and VGG, has proven particularly successful, allowing for excellent accuracy with smaller datasets and minimizing the need for lengthy training from the start. Image segmentation, which is frequently used in conjunction with CNN architectures, aids in identifying particular sections of damaged plants, increasing model accuracy. Various data augmentation strategies, such as random rotations and brightness modifications, are frequently employed to increase model generalization and reduce overfitting, particularly with little data. Furthermore, CNNs like as AlexNet, VGG, and Inception are frequently used in frameworks such as TensorFlow and Keras, allowing researchers to fine-tune models for specific plant diseases. Overall, these developments show that CNNs are successful in automating plant disease identification, providing higher accuracy and speed than previous approaches, particularly when combined with rigorous data processing and transfer learning techniques.

feature representations, although both the DCNN and AlexNet outperformed traditional classifiers.

In this section, we will discuss all the approaches we tried for image preprocessing in attempt to enhance our results by improving our findings and allowing our model to handle images more efficiently. This section will also explore the various modeling approaches we evaluated before selecting the most suitable one to pursue.

[illegible][illegible]



Figure 3 – A sample of the images we are using

2.1 Preprocessing techniques

2.1.1 Image Transformation

This was the first preprocessing approach we implemented in which we designed a code to process an image and detect brownish areas within it, which in our dataset represents the infected area, using the Canny edge detector to overlay the edges detected on the original image.

Firstly, the image is loaded and converted from the default BGR color format to RGB for proper visualization. It is then transformed into the HSV (Hue, Saturation, Value) color space, which is more effective in our case since we will use color-based segmentation.

We defined a range for detecting brownish colors in HSV. This range consists of a lower bound of [10, 50, 50] and an upper bound of [30, 255, 200]. Using these bounds, a binary mask is created that highlights all pixels within the specified brownish range. Morphological operations, specifically a closing operation with a rectangular kernel, are applied to clean the mask by removing noise and filling small holes.

The regions identified as brown are then highlighted in the original image by performing a bitwise AND operation between the image and the mask. Simultaneously, edges in the grayscale version of the original image are detected using the Canny edge detector, which identifies boundaries with significant intensity changes, signifying the infected area of the plant.

To visually combine the results, the edges are converted to an RGB format and overlaid on the brown-highlighted image using a weighted sum. This creates a composite result where both brownish regions and edges are visible.

2.1.2 Equalization

We used equalization as the second method of preprocessing the images since it is used to enhance the contrast of images to ensure that important features are more distinguishable. This improves the quality and uniformity of input data, helping the model better recognize patterns and reducing sensitivity to variations in lighting or exposure across different images. By standardizing brightness and contrast, the model can focus on meaningful features rather than being distracted by inconsistencies in the data.

Our code processes an image to improve its contrast using histogram equalization, which redistributes the brightness levels in an image for better visibility. We begin by loading an image file originally read in BGR format. Since most visualization tools use the RGB format, the image is converted from BGR to RGB.

Next, the code splits the image into its three color channels: Red, Green, and Blue. Each channel is processed individually using histogram equalization. This technique enhances the contrast by spreading out the intensity distribution of the pixel values, ensuring better differentiation between light and dark areas.

After equalizing the individual channels, they are merged back together to form a single image. The result is an enhanced version of the original image where the contrast of each color channel has been adjusted. To make the enhanced image suitable for visualization, it is converted back to RGB format.

2.1.3 Shadow Removal

Another attempt to preprocess the images before supplying them to the model was trying to remove the shadows from them. This attempt was to enhance image uniformity for better processing and analysis. Shadow removal is essential in preprocessing as shadows can obscure important details and introduce variability in image data.

The process begins by reading the image in BGR format and converting it into the $YCbCr$ color space. This is done because $YCbCr$ separates brightness from chromatic information, C_b and C_r , making it easier to analyze shadows based on intensity. A copy of the $YCbCr$ image is created to serve as a binary mask for classifying shadow and non-shadow regions.

The mean and standard deviation of the Y channel, which is the brightness, are then calculated to determine a threshold for shadow classification. Pixels with intensity below the threshold, which is defined as the mean minus a fraction of the standard deviation, are classified as shadow pixels and marked as white in the binary mask, while others are marked as black.

After that, morphological erosion is applied to the mask to refine shadow detection and reduce misclassified pixels. The algorithm then calculates the sum and count of pixel intensities separately for shadow and non-shadow areas. Using these values, it computes the average brightness in both regions and the intensity difference between them.

To normalize the shadowed areas, the intensity difference and a ratio based on the averages are applied to the shadowed pixels. This adjustment ensures that the shadowed regions match the brightness of the lit areas, effectively removing shadows.

2.1.4 Intrinsic Image

This was our final preprocessing approach, and it was aimed at reducing the impact of lighting variations by separating object colors from illumination to ensure that the model focuses on the true features of the image in attempts of improving consistency and robustness in analysis.

This code is designed to preprocess images using Retinex theory, which separates an image into two components: reflectance, which is the true color of objects, and illumination, which represents the lighting conditions. This decomposition helps enhance image features by reducing the impact of lighting variations, making it suitable for tasks such as object recognition and shadow removal.

The core function we implemented takes an image and performs the decomposition. The image is first loaded and converted to RGB format. It is normalized to the range [0, 1] and transformed into the logarithmic domain to simplify calculations. Illumination is estimated using a Gaussian blur, which smooths the image, while reflectance is calculated by subtracting the smoothed illumination from the logarithmic image. Both components are then converted back to their original scale for further use.

2.2 Modelling techniques

For modelling we tried two different approaches using a pretrained VGG16 model and our own constructed deep convolutional neural network.

2.2.1 VGG16

VGG16 is a convolutional neural network (CNN) model that we utilized as part of our approach to classify plant diseases from images. As a pretrained model on the ImageNet dataset, it has already learned to recognize a wide range of general image features, which makes it an excellent choice for transfer learning.

We used VGG16 to process plant images by extracting meaningful features through its convolutional layers, which are well-suited for identifying patterns and textures in leaves and stems. These extracted features are then used to classify the images into various disease categories. By leveraging its pretrained weights, we were able to focus on fine-tuning the model for plant disease classification, reducing the need for extensive training from scratch.

2.2.2 CNN

Our deep convolutional neural network (CNN) is specifically designed for multi-class image classification tasks, with the capability to classify images into 38 distinct categories. The model processes input images of dimensions $128 \times 128 \times 3$ representing RGB color channels, and is implemented using the TensorFlow Keras Sequential API. Its architecture comprises a series of convolutional blocks, followed by fully connected layers to ensure comprehensive feature extraction and accurate classification.

The convolutional architecture consists of five hierarchical blocks, each containing two convolutional layers with 3×3 filters. The number of filters increases progressively from 32 to 512 across the blocks, enabling the model to extract features of increasing complexity. Initial layers focus on learning basic patterns such as edges and textures, while deeper layers capture more intricate features. Max-pooling layers follow each block to reduce the spatial dimensions of the feature maps, optimizing computational efficiency and minimizing the risk of overfitting.

To further mitigate overfitting, dropout layers are incorporated at key stages of the network. A dropout rate of 0.25 is applied after the convolutional blocks, and a rate of 0.4 is used before the dense layers. These layers deactivate a fraction of neurons during training, enhancing the model's generalization capabilities.

The network concludes with a dense section, where extracted features are flattened into a single vector and passed through a fully connected layer with 1500 units and ReLU activation. The final layer employs a softmax activation function to produce a probability distribution over the 38 target categories. This architecture demonstrates robustness and efficiency in multi-class classification tasks, providing an effective balance between feature extraction and computational demands.

3. Results

The first thing we did was to design a code that creates a data augmentation pipeline to enhance the diversity of

training images and improve the model's ability to generalize. The code applies random transformations such as flipping, rotating, zooming, and adjusting brightness and contrast to simulate real-world variations.

Additionally, it normalizes pixel values to a range of [0, 1] for numerical stability during training. We did these augmentations to make the model more robust to variations in input data, and to enrich the training data by introducing variability, helping the model become more robust to real-world image variations.

Following those image generalization transformations, we tried our two different modelling approaches, the self-constructed CNN, which yielded an accuracy of 97.35%, and the pretrained VGG16 model, which resulted in an 89.23% accuracy.

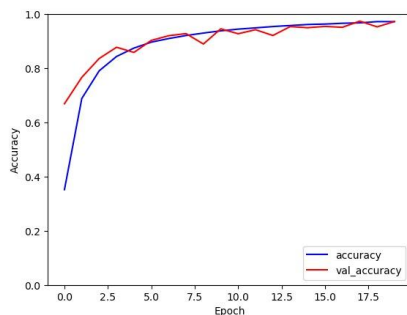


Figure 4 – Accuracy of the CNN model

In contrast to our expectations regarding the preprocessing's effectiveness, the results turned out to be more robust and accurate when we did not apply any preprocessing technique and just the image generalization transformations for various reasons.

3.1 Image Transformation

Below are the results of this technique.

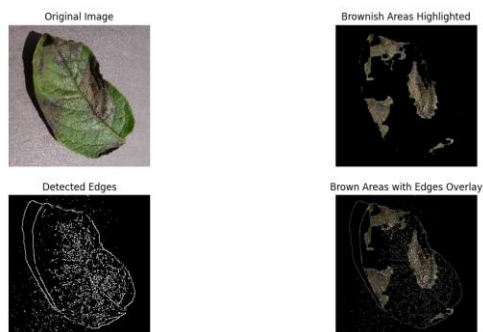


Figure 5 – Image Transformation Results



Figure 6 – Image Transformation Results



Figure 7 – Image Transformation Results

Despite the success of this technique on a multitude of images, it posed some problems in some corner cases. For example, some of the infected parts of the plants were not necessarily brown in color which resulted in them going undetected by this preprocessing approach. Another problem was that the illness in some cases was represented by a cut in the plant or a removal of a part of its body, this also went undetected by that technique. Finally, some images already had some brown details in the background or across it which were not related to the illness and were detected as infected areas by the processing technique. All these cases resulted in distortions to our model, which led to a decrease in its accuracy.

To address these issues, incorporating advanced segmentation techniques, such as semantic segmentation models or region-based methods, could improve the detection of infected areas by focusing on plant-specific features rather than relying solely on color-based preprocessing.

3.2 Equalization

Below are the results of this technique.



Figure 7 – Image Equalization Results



Figure 8 – Image Equalization Results



Figure 9 – Image Equalization Results

As for this approach, although it produced more enhanced images, it decreased the accuracy of the model and caused it to overfit due to the change in color in multiple regions of the image.

To address this limitation, applying adaptive histogram equalization selectively or combining it with other preprocessing methods that preserve color consistency could enhance image quality without compromising model accuracy.

3.3 Shadow Removal

Below are the results of this technique.

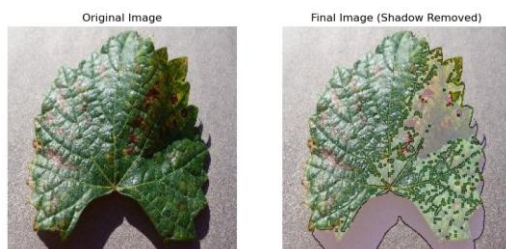


Figure 10 – Shadow Removal Results

This approach caused a major problem since in many cases, there were shadows on the plant itself, not just in the background or surroundings. When the shadow was extracted from the images in these cases it produced a misleading, unreliable image, which produced inaccurate results when given to our model.

A potential solution to this issue is to incorporate advanced shadow detection techniques that can differentiate between shadows on the plant and those in the background, ensuring more reliable preprocessing and preserving the integrity of the input images.

3.4 Intrinsic image

Below are the results of this technique.



Figure 11 – Intrinsic Image Results

This approach manipulated the contrast of the images, which worked fine for some of the data, but failed for the majority. We tried to enhance it by fine-tuning the approach, but we could not find a fine-tuning method that suited all the images to yield correct results.

One way to address this challenge is to explore adaptive contrast enhancement methods that can dynamically adjust to the specific characteristics of each image, ensuring consistent and accurate preprocessing across the dataset.

3.5 Final remarks

We strongly believe that if we perform further research to tailor our preprocessing approaches more, the yielded results might be more accurate and reliable, so this for us is a future area of improvement. Also, another further enhancement would be applying object detection using YOLO to only detect the plant we are examining and remove the background to allow our dataset to be more reliable to training and produce more robust results.

To conclude, our cutting edge is that we used our own custom-built CNN model, achieving an impressive accuracy of 97.35% without relying on any pretrained models. This decision significantly reduced both storage requirements and computational complexity, making the process more efficient. In contrast to our initial

experiments with VGG16 and insights from the literature reviews that revealed a heavy reliance on pretrained models in previous research.

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Statement of Individual Contribution

- Literature Review and Research: Malak & Merna
- Data Preprocessing and Augmentation: Mohanad & Mahmoud & Hadj
- Model Development: Mohanad
- Implementation of Evaluation Metrics: Mahmoud
- Experimentation and Troubleshooting: Mohanad & Hadj
- Technical Implementation and Debugging: Mohanad & Hadj & Mahmoud
- Documentation and Report Writing: Malak & Merna
- Results Interpretation and Future Recommendations: Malak & Merna

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